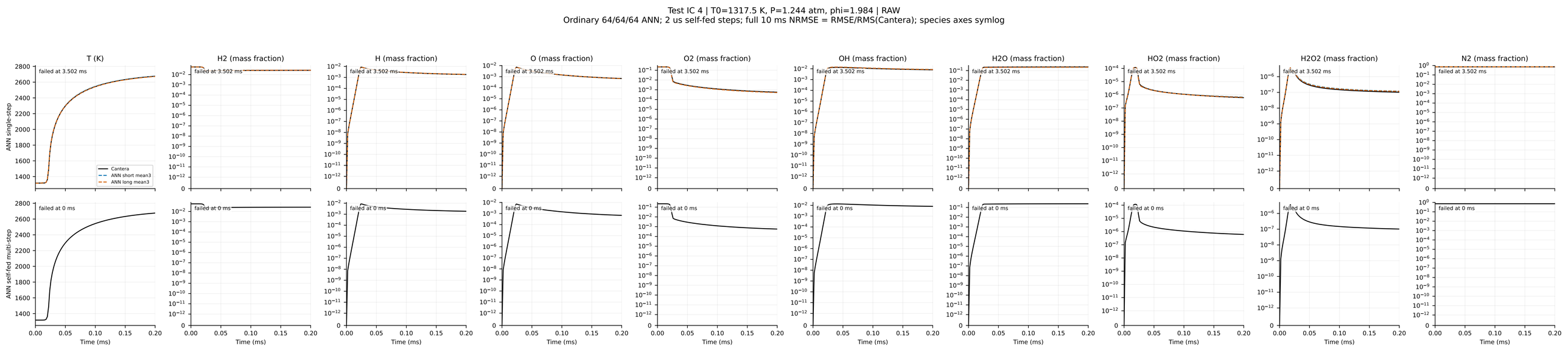
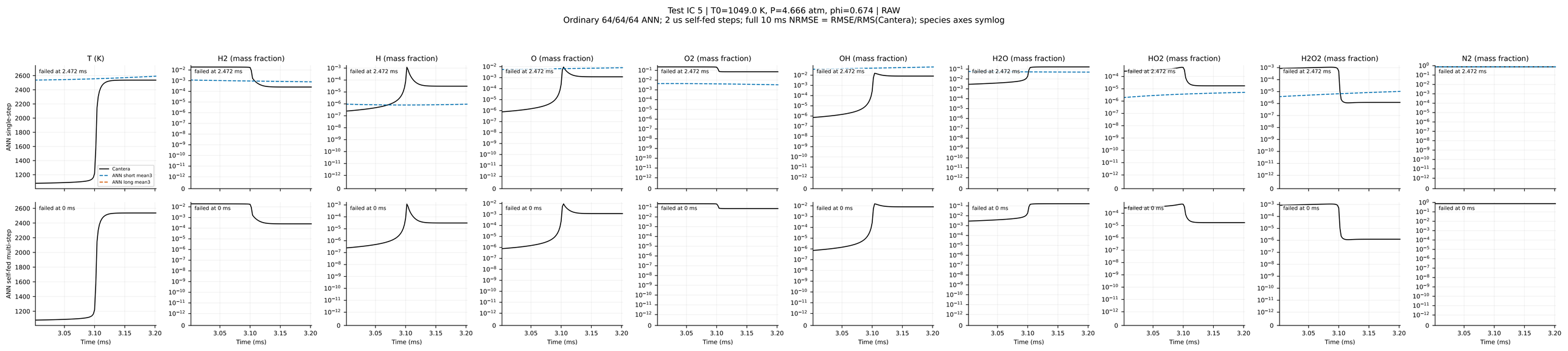
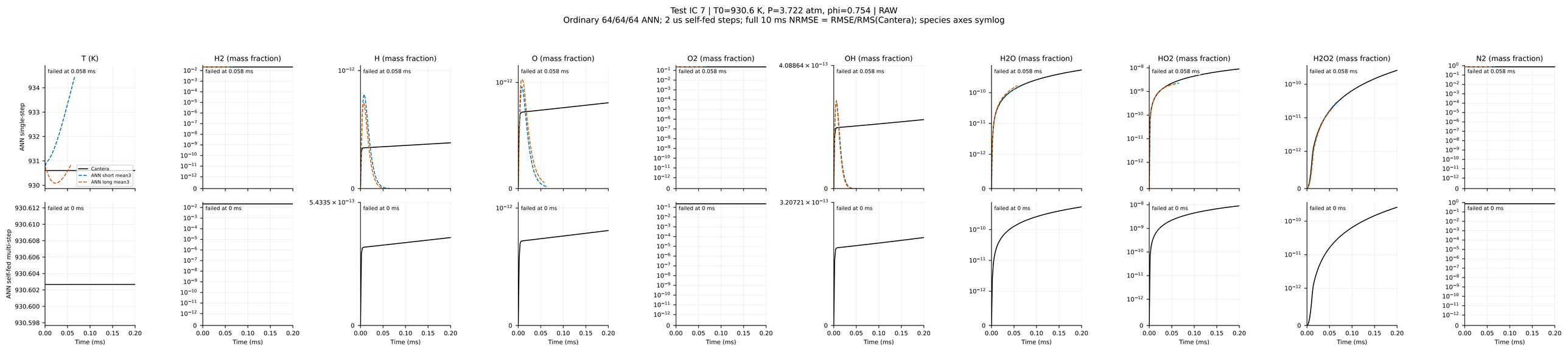
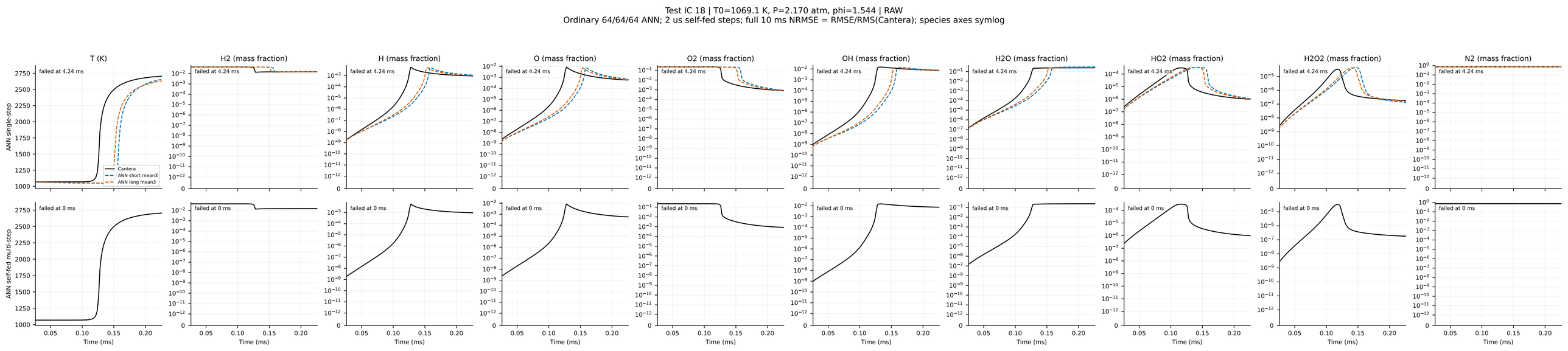
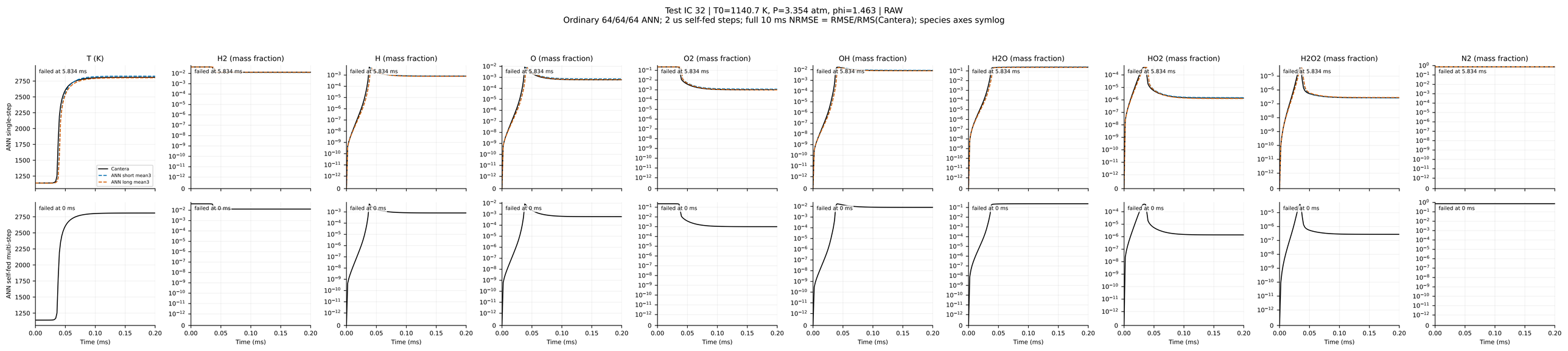
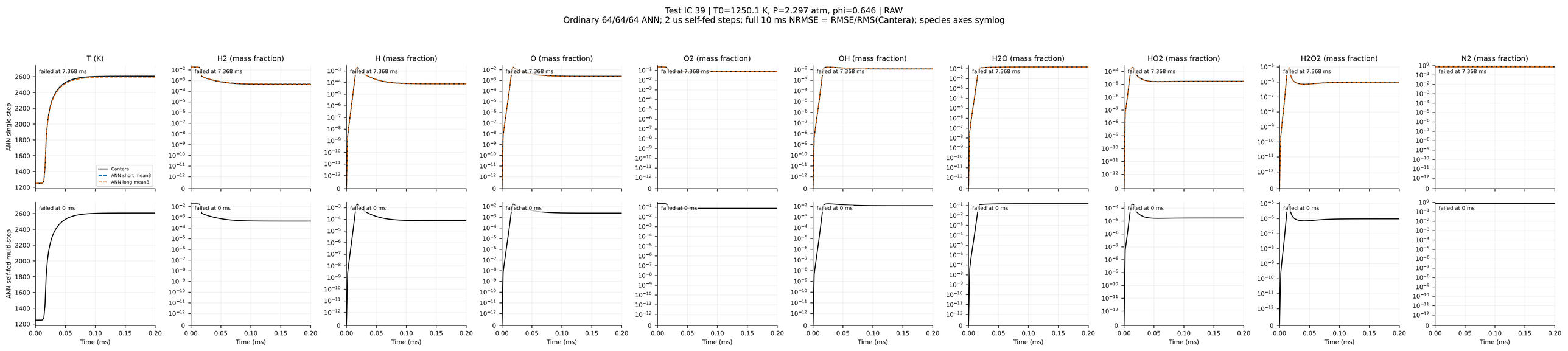

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
Cantera
ANN short mean3
ANN long mean3
failed at 3.502 ms
failed at 0 ms
Time (ms)

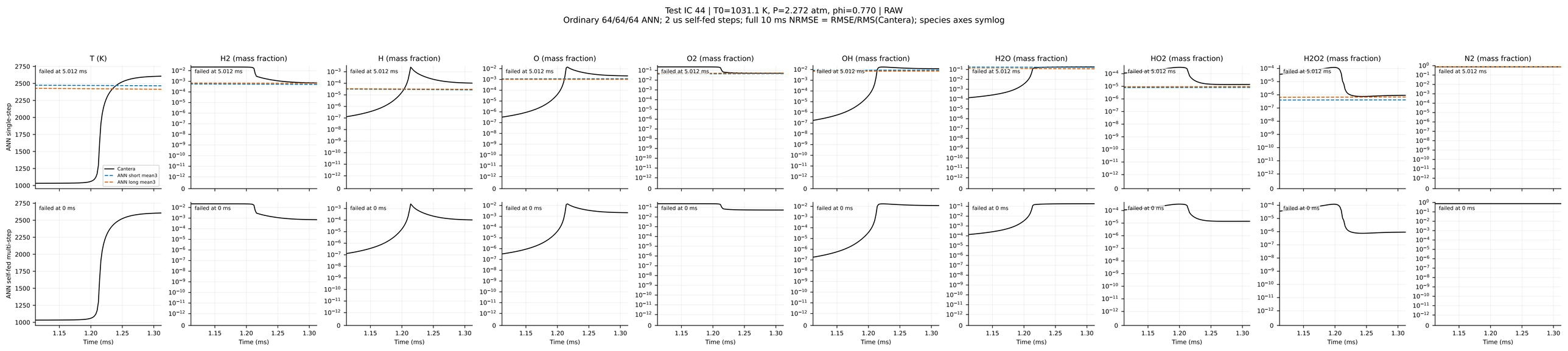

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 2.472 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

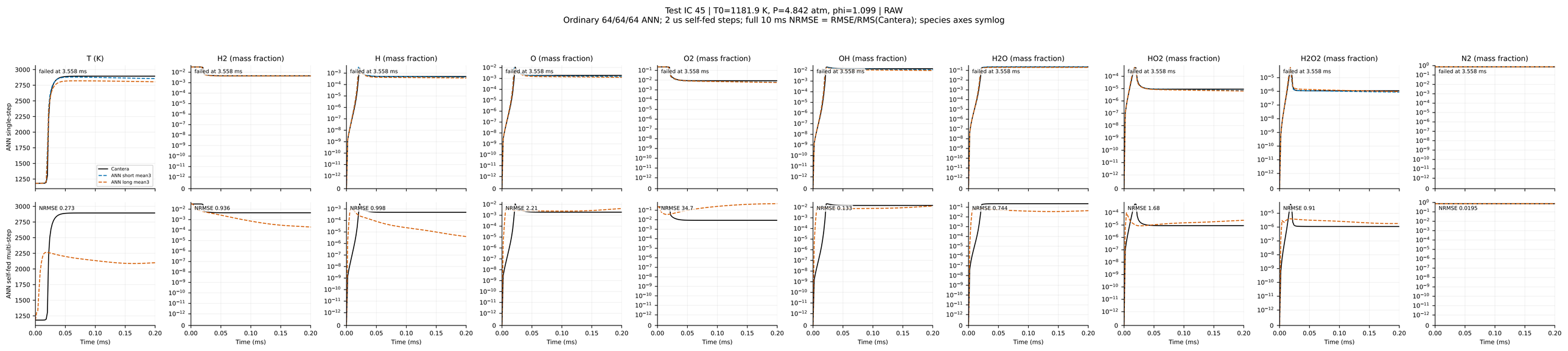

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 0.058 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

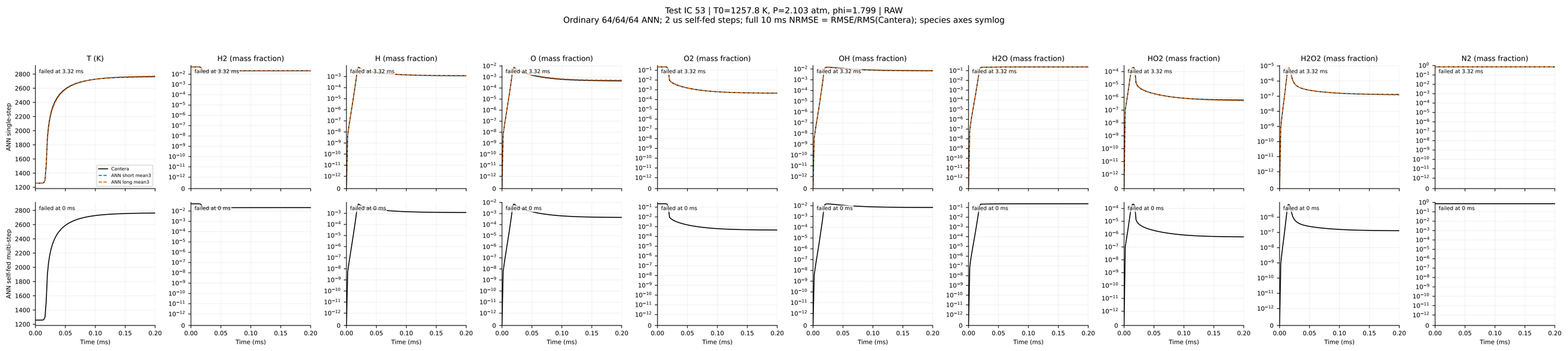

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 4.24 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

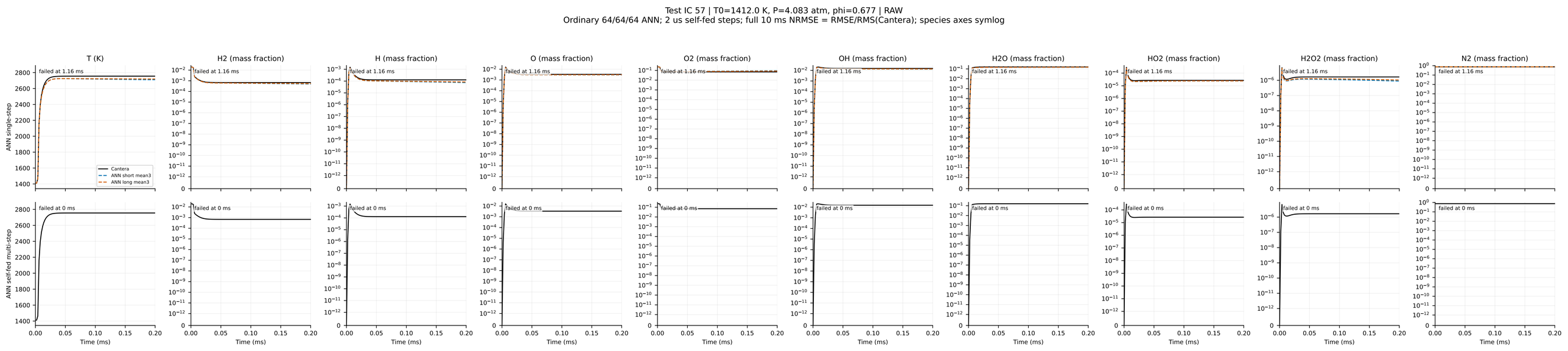

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 5.834 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

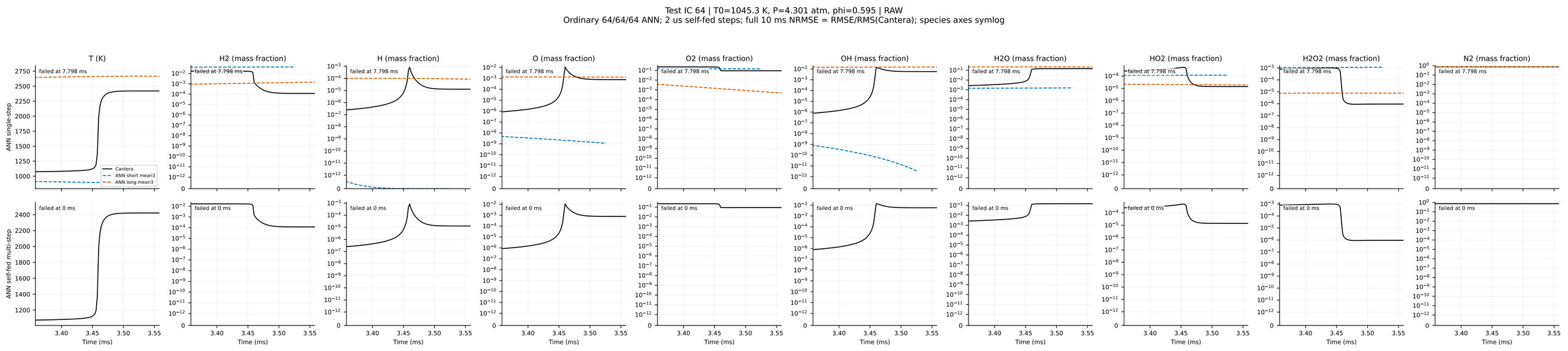

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 7.368 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

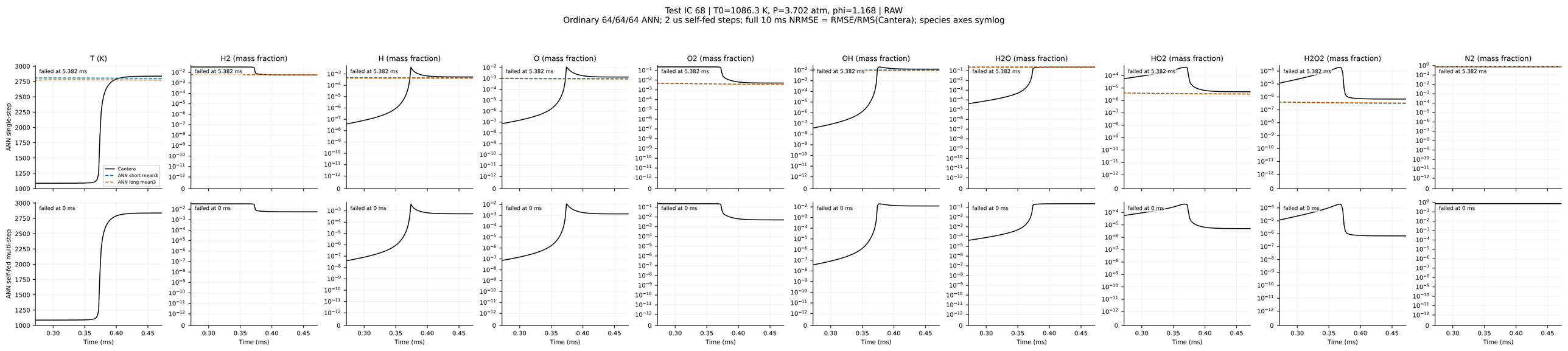

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 5.012 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

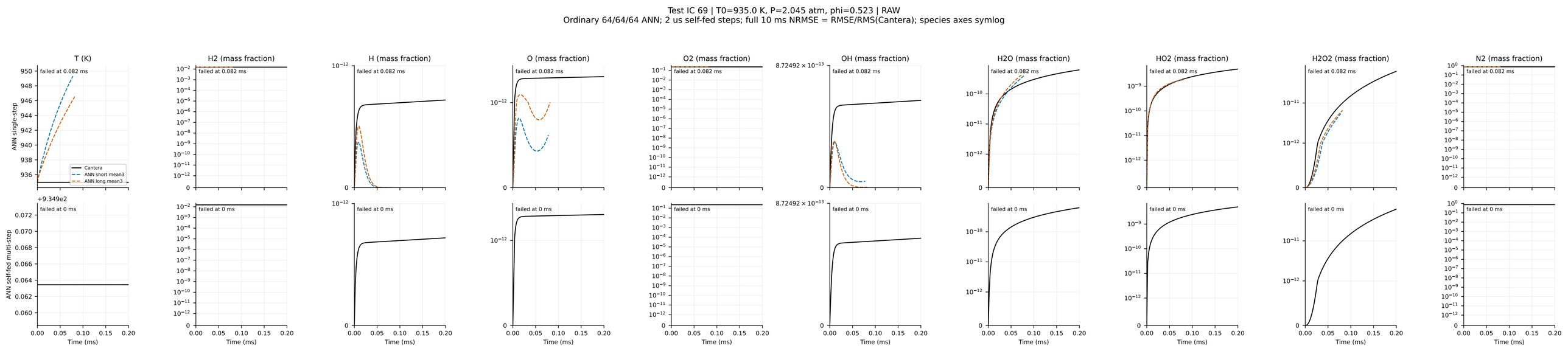

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
Cantera
ANN short mean3
ANN long mean3
failed at 3.558 ms
NRMSE 0.273
NRMSE 0.936
NRMSE 0.998
NRMSE 2.21
NRMSE 34.7
NRMSE 0.133
NRMSE 0.744
NRMSE 1.68
NRMSE 0.91
NRMSE 0.0195
Time (ms)

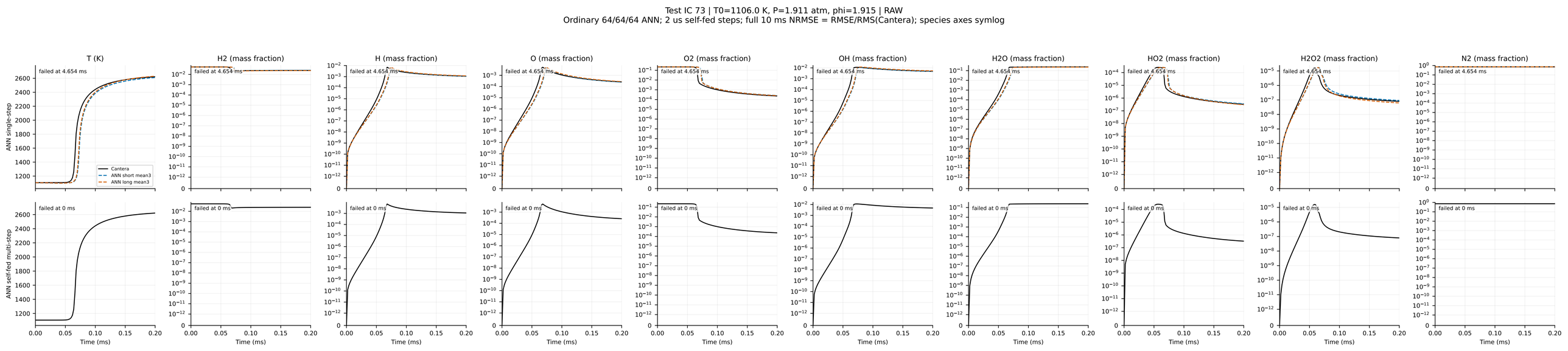

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 3.32 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

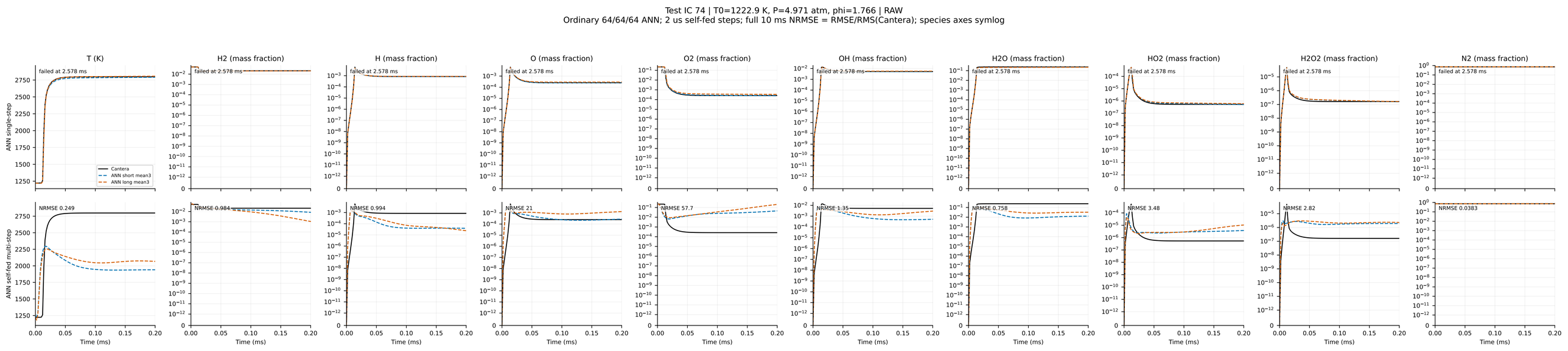

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 1.16 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

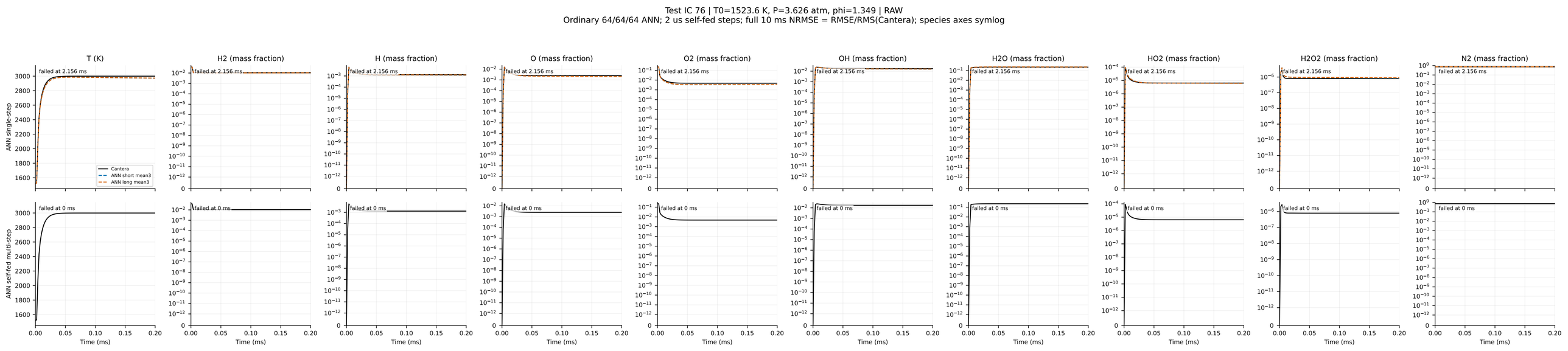

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 7.798 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

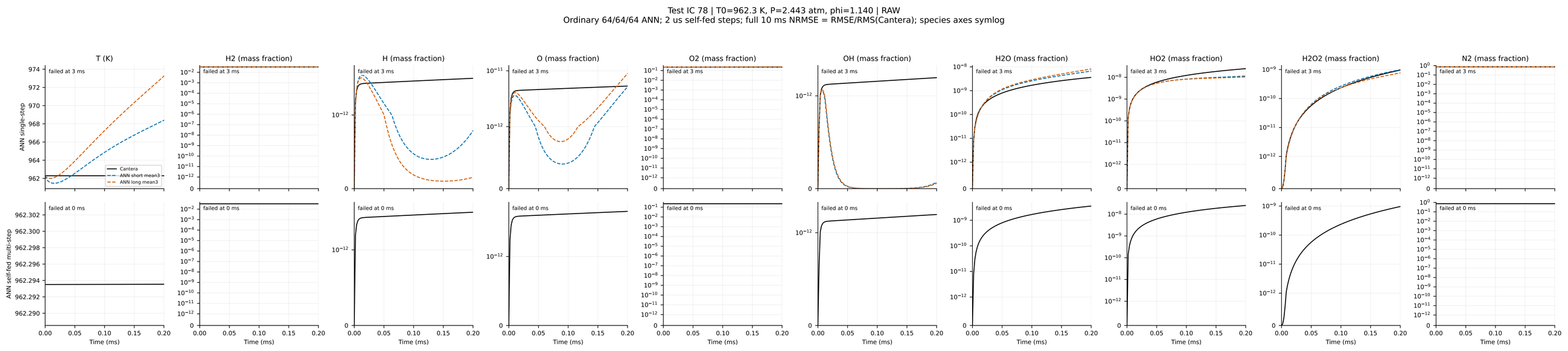

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 5.382 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

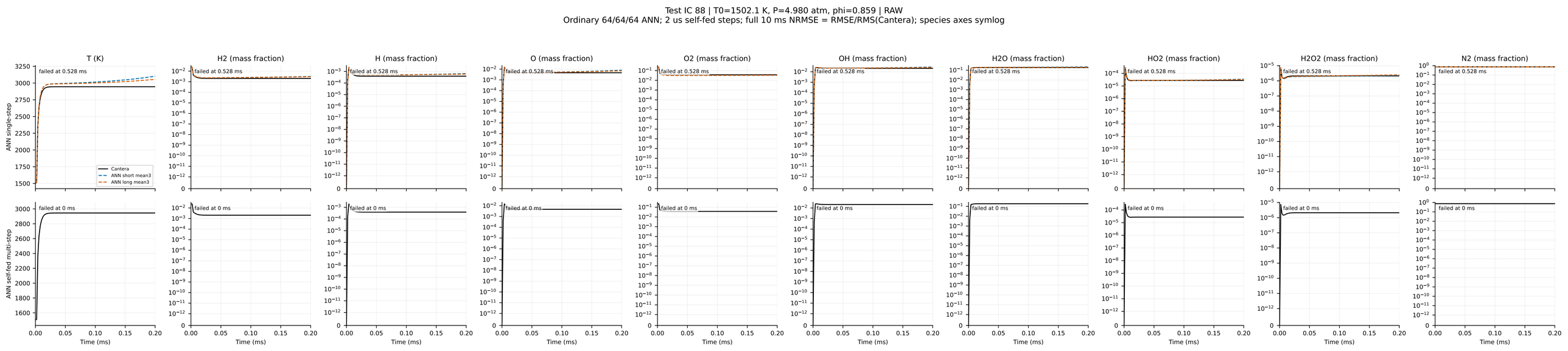

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
Cantera
ANN short mean3
ANN long mean3
failed at 0.082 ms
failed at 0 ms
+9.349e2
8.72492 × 10⁻¹³
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 4.654 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 2.578 ms
Cantera
ANN short mean3
ANN long mean3
NRMSE 0.249
NRMSE 0.984
NRMSE 0.994
NRMSE 21
NRMSE 57.7
NRMSE 1.35
NRMSE 0.758
NRMSE 3.48
NRMSE 2.82
NRMSE 0.0383
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
Cantera
ANN short mean3
ANN long mean3
failed at 2.156 ms
failed at 0 ms
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 3 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 0.528 ms
failed at 0 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)